Lecture Notes

Lecture 4

Matrices of Linear Maps; Change of Basis

Lecture 3 ended with a liberating fact: a linear map is completely determined by what it does to a basis. If we also fix a basis of the target space, each of those output vectors becomes a column of coordinates—and stacking the columns gives a *matrix*. This lecture turns the informal dictionary "linear map $\leftrightarrow$ matrix" into a precise isomorphism. Three payoffs follow. First, the rule for multiplying matrices is *forced* on us: it is exactly what composition of linear maps demands. Second, abstract spaces of linear maps turn out to be ordinary matrix spaces. Third, when we change basis the same operator acquires a new matrix related to the old one by *similarity* $A' = P^{-1}AP$.

The basis-dependence is the whole point for physics. In the finite/discrete core, choosing a basis chooses a coordinate representation; discretized position, momentum, and energy models are useful examples. Continuous position or momentum representations and continuous spectral measures require the Hilbert-space machinery of Lectures 13–14. The later finite lectures choose bases in which an operator's matrix is as simple as possible.

Learning objectives

By the end of this lecture you will be able to:

- Build the matrix $\mathcal{M}(T)$ in chosen bases and use $[Tv] = \mathcal{M}(T)[v]$.
- Explain why composition of maps is matrix multiplication.
- Decide invertibility and classify spaces up to isomorphism by dimension.
- Change basis and identify trace and determinant as similarity invariants.

1 The matrix of a linear map

Fix bases $\mathcal{B} = (v_1, \dots, v_n)$ of V and $\mathcal{C} = (w_1, \dots, w_m)$ of W.

Definition 4.1 (Matrix of a linear map). For $T \in \mathcal{L}(V, W)$, the *matrix* of T with respect to $\mathcal{B}$ and $\mathcal{C}$ is the $m \times n$ matrix $\mathcal{M}(T) = (A_{jk})$ whose k-th column holds the $\mathcal{C}$-coordinates of Tv_k:

$$Tv_k = A_{1k}w_1 + \cdots + A_{mk}w_m = \sum_{j=1}^{m} A_{jk}\, w_j.$$

When the bases are not clear from context we write $\mathcal{M}\big(T, \mathcal{B}, \mathcal{C}\big)$.

In words: *the k-th column is T applied to the k-th basis vector, written in coordinates.* For a map $\mathbb{F}^n \to \mathbb{F}^m$ with the standard bases, the k-th column is simply Te_k.

Example 4.2 (A map $\mathbb{F}^2 \to \mathbb{F}^3$). Let $T(x, y) = (x+3y,\ 2x+5y,\ 7x+9y)$. Since $T(1, 0) = (1, 2, 7)$ and $T(0, 1) = (3, 5, 9)$,

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

The columns are the images of the standard basis vectors. ◁

Example 4.3 (Rotation of the plane). Rotation R_θ of $\mathbb{R}^2$ by an angle θ sends $e_1 \mapsto (\cos\theta, \sin\theta)$ and $e_2 \mapsto (-\sin\theta, \cos\theta)$, so

$$\mathcal{M}(R_\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Composing rotations multiplies their matrices, and the trigonometric angle-addition formulae fall out of $\mathcal{M}(R_\alpha)\mathcal{M}(R_\beta) = \mathcal{M}(R_{\alpha+\beta})$. ◁

Example 4.4 (Differentiation on $\mathcal{P}_3$). Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be $Dp = p'$, with the standard basis $1, t, t^2, t^3$. From $D1 = 0$, $Dt = 1$, $Dt^2 = 2t$, $Dt^3 = 3t^2$,

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Each column records a derivative in coordinates; the superdiagonal carries the exponents $1, 2, 3$. ◁

Example 4.5 (The matrix depends on the basis). On $\mathcal{P}_2(\mathbb{R})$, the standard basis gives $\mathcal{M}(D) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, but the basis $\mathcal{B}' = (1,\ 1+t,\ 1+t+t^2)$ gives a different matrix. Computing and re-expanding in $\mathcal{B}'$: $D1 = 0$; $D(1+t) = 1$ has $\mathcal{B}'$-coordinates $(1, 0, 0)$; and $D(1+t+t^2) = 1+2t$ has coordinates $(-1, 2, 0)$ (solve $a+b+c = 1,\ b+c = 2,\ c = 0$). Hence

$$\mathcal{M}(D, \mathcal{B}') = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Same operator, different matrix; §4 explains exactly how the two relate. ◁

The defining property propagates from basis vectors to all vectors. Writing $[v]_\mathcal{B}$ for the coordinate column of v, we get the identity that makes matrices useful for computation.

Theorem 4.6 (A matrix acts on coordinates). For every $v \in V$,

$$[Tv]_C = \mathcal{M}(T)\,[v]_\mathcal{B}.$$

Proof. Write $v = \sum_k b_k v_k$, so $[v]_\mathcal{B} = (b_1, \ldots, b_n)$. By linearity,

$$Tv = \sum_k b_k\, Tv_k = \sum_k b_k \sum_j A_{jk} w_j = \sum_j \Big(\sum_k A_{jk} b_k\Big) w_j.$$

Thus the j-th C-coordinate of Tv is $\sum_k A_{jk} b_k$. These are precisely the entries of the matrix product in the theorem. ■

This is the content of the square in Figure 1: taking coordinates and applying T commute with multiplying by $\mathcal{M}(T)$.

Example 4.7 (Computing Tv through the matrix). Take D on $\mathcal{P}_3(\mathbb{R})$ (Example 4.4) and $p(t) = 2 + t - t^2 + 4t^3$, so $[p] = (2, 1, -1, 4)$. Then

$$[Dp] = \mathcal{M}(D)\,[p] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 12 \\ 0 \end{pmatrix},$$

that is $Dp = 1 - 2t + 12t^2$, matching direct differentiation. The matrix has turned calculus into arithmetic. ◁

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "{[\,\cdot\,]_{\mathcal{B}}}"'] & W \arrow[d, "{[\,\cdot\,]_{\mathcal{C}}}"] \\ \mathbb{F}^n \arrow[r, "\mathcal{M}(T)"'] & \mathbb{F}^m \end{tikzcd}$$

Figure 1: The matrix is the map in coordinates: descending and then multiplying by $\mathcal{M}(T)$ gives the same result as applying T and then descending.

2 Composition becomes matrix multiplication

Matrices are added and scaled entrywise. With those definitions, $\mathcal{M}$ respects the linear structure.

Proposition 4.8 ($\mathcal{M}$ is linear). For $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, $\mathcal{M}(S+T) = \mathcal{M}(S)+\mathcal{M}(T)$ and $\mathcal{M}(aT) = a\,\mathcal{M}(T)$.

Proof. The k-th column of $\mathcal{M}(S+T)$ holds the $\mathcal{C}$-coordinates of $(S+T)v_k = Sv_k + Tv_k$, which by linearity of coordinates is the sum of the k-th columns of $\mathcal{M}(S)$ and $\mathcal{M}(T)$. The scalar case is identical. ∎

The deeper question is what corresponds to *composition*—and its answer dictates how matrices must multiply.

Theorem 4.9 (Composition → multiplication). Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, with bases fixed on all three spaces. Then $\mathcal{M}(ST) = \mathcal{M}(S)\,\mathcal{M}(T)$, where the product of an $m \times p$ matrix $B = (B_{ij})$ and a $p \times n$ matrix $A = (A_{jk})$ is the $m \times n$ matrix with entries

$$(BA)_{ik} = \sum_{j=1}^{p} B_{ij}A_{jk}. \tag{4.1}$$

Proof. Let $\mathcal{B} = (u_1, \ldots, u_n)$ be the basis of U, with $\mathcal{M}(T) = (A_{jk})$ and $\mathcal{M}(S) = (B_{ij})$. Applying T then S to the k-th basis vector,

$$(ST)u_k = S\Big(\sum_j A_{jk}v_j\Big) = \sum_j A_{jk}\,Sv_j = \sum_j A_{jk}\sum_i B_{ij}w_i = \sum_i\Big(\sum_j B_{ij}A_{jk}\Big)w_i.$$

The i-th coordinate of $(ST)u_k$—that is, the (i,k) entry of $\mathcal{M}(ST)$—is $\sum_j B_{ij}A_{jk}$, which is precisely (4.1). ∎

So the "row-times-column" rule for matrix products is not an arbitrary convention: it is the only rule that makes matrices represent composition. Every property of matrix multiplication—associativity, distributivity, failure to commute—is inherited from the corresponding property of composing maps.

Example 4.10 ($\mathcal{M}(D^2) = \mathcal{M}(D)^2$). With D on $\mathcal{P}_3(\mathbb{R})$ as in Example 4.4, the operator D^2 sends $1, t, t^2, t^3$ to $0, 0, 2, 6t$. Multiplying the matrix of D by itself reproduces exactly this:

$$\mathcal{M}(D)^2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D^2),$$

the nonzero band having shifted up by one and picked up the products 2, 6 of consecutive exponents. ◁

Example 4.11 (Multiplication realizes composition). Let $T(x,y) = (x+y,\ y)$ and $S(x,y) = (2x,\ x-y)$ on $\mathbb{R}^2$, with matrices $\mathcal{M}(T) = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\mathcal{M}(S) = \left(\begin{smallmatrix}2&0\\1&-1\end{smallmatrix}\right)$. Composing, $(ST)(x,y) = S(x+y,y) = (2x+2y,\ x)$, whose matrix is $\left(\begin{smallmatrix}2&2\\1&0\end{smallmatrix}\right)$. The product $\mathcal{M}(S)\mathcal{M}(T) = \left(\begin{smallmatrix}2&0\\1&-1\end{smallmatrix}\right)\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right) = \left(\begin{smallmatrix}2&2\\1&0\end{smallmatrix}\right)$ agrees, exactly as Theorem 4.9 requires. ◁

The map $\mathcal{M}$ from $\mathcal{L}(V,W)$ to the matrix space $\mathbb{F}^{m\times n}$ is linear, injective (a map with zero matrix kills every basis vector, hence is 0), and surjective (any matrix prescribes images of the basis vectors, which define a unique linear map, as in Lecture 3). Thus it is an isomorphism, and we can finally record the dimension promised in Lecture 3.

> **Corollary 4.12 ($\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$).** $\mathcal{M}$ is an isomorphism from $\mathcal{L}(V,W)$ onto $\mathbb{F}^{m\times n}$. In particular $\dim \mathcal{L}(V,W) = (\dim V)(\dim W)$.

> **Proposition 4.13 (Matrix rank is map rank).** For any choice of bases, dim range T equals the column rank of $\mathcal{M}(T)$, the dimension of the span of its columns.

Proof. The coordinate isomorphism $[\,\cdot\,]_C \colon W \cong \mathbb{F}^m$ carries range $T = \operatorname{span}(Tv_1,\dots,Tv_n)$ onto the span of the vectors $[Tv_k]_C$, which are precisely the columns of $\mathcal{M}(T)$. An isomorphism preserves dimension, so dim range T equals the column rank. ■

Remark 4.14. This also connects map rank with Gaussian elimination. Reduce a matrix A to reduced row-echelon form $R = EA$, where E is invertible. Row operations preserve the row space, and $EAx = 0 \iff Ax = 0$, so they preserve all linear dependencies among the columns. If R has r pivots, its r nonzero rows are independent, while its pivot columns are independent and every nonpivot column is a combination of them. Thus both row rank and column rank of R equal r; the preserved row space and column dependencies give the same equality for A. Consequently elimination computes the basis-independent map rank.

Example 4.15 (Dimension of a map space). Hence $\dim \mathcal{L}(\mathbb{R}^3,\mathbb{R}^2) = 6$, with a basis given by the six maps whose matrices carry a single 1 in one of the six entries and zeros elsewhere. ◁

3 Invertibility and isomorphisms

> **Definition 4.16 (Invertible map, isomorphism).** $T \in \mathcal{L}(V,W)$ is *invertible* if there is $S \in \mathcal{L}(W,V)$ with $ST = I_V$ and $TS = I_W$; such S is unique, written T^{-1}. An invertible linear map is an *isomorphism*, and V, W are *isomorphic*, $V \cong W$, if one exists.

A linear map is invertible exactly when it is bijective, as the next proposition records; the decisive fact for finite-dimensional spaces is then that a single number—the dimension—decides isomorphism.

> **Proposition 4.17 (Invertible = bijective).** $T \in \mathcal{L}(V,W)$ is invertible if and only if it is both injective and surjective, and then its inverse function is automatically linear.

Proof. An invertible map has a two-sided inverse, so it is a bijection. Conversely, if T is a bijection let T^{-1} be its inverse function. Writing $w_i = Tv_i$, for $a \in \mathbb{F}$ we have $T^{-1}(w_1 + aw_2) = T^{-1}\big(T(v_1 + av_2)\big) = v_1 + av_2 = T^{-1}w_1 + a\,T^{-1}w_2$, so T^{-1} is linear; it is a genuine two-sided inverse by construction. ■

Theorem 4.18 (Dimension classifies spaces). Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

Proof. If $T: V \to W$ is an isomorphism then $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = W$, so by rank–nullity $\dim V = \dim \operatorname{range} T = \dim W$. Conversely, if $\dim V = \dim W = n$, pick bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_n$ of W and let T be the linear map with $Tv_k = w_k$ (Lecture 3). It is surjective because the w_k span W and injective because they are independent (so $\operatorname{null} T = \{0\}$); hence it is an isomorphism. ∎

Corollary 4.19 (Coordinates are an isomorphism). If $\dim V = n$, the coordinate map $v \mapsto [v]_{\mathcal{B}}$ is an isomorphism $V \cong \mathbb{F}^n$. Thus every n-dimensional space over $\mathbb{F}$ is a copy of $\mathbb{F}^n$; e.g. $\mathcal{P}_m(\mathbb{F}) \cong \mathbb{F}^{m+1}$ and $M_{m\times n}(\mathbb{F}) \cong \mathbb{F}^{mn}$.

Example 4.20 (An explicit isomorphism). The space $\mathrm{Sym}_2(\mathbb{R})$ of symmetric 2×2 real matrices and $\mathbb{R}^3$ both have dimension 3, hence are isomorphic. An explicit isomorphism is $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right) \mapsto (a, b, c)$, which is exactly the coordinate map for the basis $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Spaces that look quite different are, for the purposes of linear algebra, the same once their dimensions match. ◁

Example 4.21 (Transporting a question to $\mathbb{F}^n$). Do $1+t,\ t+t^2,\ 1+t^2$ form a basis of $\mathcal{P}_2(\mathbb{R})$? Under the coordinate isomorphism $\mathcal{P}_2(\mathbb{R}) \cong \mathbb{R}^3$ (standard basis $1, t, t^2$) they become $(1,1,0), (0,1,1), (1,0,1)$, and

$$\det\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 2 \neq 0$$

shows these are independent in $\mathbb{R}^3$. An isomorphism preserves independence and spanning, so the three polynomials are a basis of $\mathcal{P}_2(\mathbb{R})$—an abstract question settled by one numerical determinant. ◁

In matrix terms, T is invertible if and only if $\mathcal{M}(T)$ is an invertible square matrix, and then $\mathcal{M}(T^{-1}) = \mathcal{M}(T)^{-1}$—immediate from Theorem 4.9 applied to $T^{-1}T = I$.

Example 4.22 (Invertibility through the matrix). The operator $T(x,y) = (x+2y,\ 3x+4y)$ on $\mathbb{R}^2$ has matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ with $\det A = -2 \neq 0$, so T is invertible. Then $A^{-1} = \frac{1}{-2}\left(\begin{smallmatrix} 4 & -2 \\ -3 & 1 \end{smallmatrix}\right) = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$ represents T^{-1}, i.e. $T^{-1}(x,y) = \left(-2x+y,\ \frac{3}{2}x - \frac{1}{2}y\right)$. Invertibility of the operator and of its matrix are one and the same. ◁

Example 4.23 (A non-invertible operator). Differentiation D on $\mathcal{P}_3(\mathbb{R})$ is not invertible: its matrix $\left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$ has a zero column, so $\det = 0$, and indeed $\operatorname{null} D$ contains the nonzero constants. No change of basis can repair this—invertibility is a property of the operator, not of the matrix that represents it. ◁

4 Change of basis and similarity

The matrix of a map depends on the chosen bases. We now pin down exactly how it changes, for an operator $T \in \mathcal{L}(V)$ described in two bases of the same space V. Let $\mathcal{B} = (v_1, \dots, v_n)$ be the "old" basis and $\mathcal{B}' = (v_1', \dots, v_n')$ the "new" one.

Definition 4.24 (Change-of-basis matrix). The *change-of-basis matrix* P is the $n \times n$ matrix whose k-th column holds the $\mathcal{B}$-coordinates of the new basis vector v'_k. Equivalently $P = \mathcal{M}(\mathrm{id}, \mathcal{B}', \mathcal{B})$.

Because P represents the identity, Theorem 4.6 gives, for every v,

$$[v]_{\mathcal{B}} = P\,[v]_{\mathcal{B}'}. \tag{4.2}$$

P is invertible (it represents the invertible identity map), so also $[v]_{\mathcal{B}'} = P^{-1}[v]_{\mathcal{B}}$.

Example 4.25 (Transforming coordinates). In $\mathbb{R}^2$ let $\mathcal{B}$ be the standard basis and $\mathcal{B}' = \big((1,1),(1,-1)\big)$, so $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. A vector with new coordinates $[v]_{\mathcal{B}'} = (3,1)$ has old coordinates $[v]_{\mathcal{B}} = P\,(3,1) = (4,2)$, i.e. $v = (4,2)$—consistent with $3(1,1) + 1(1,-1) = (4,2)$. ◁

Theorem 4.26 (Similarity). Let $T \in \mathcal{L}(V)$ have matrices $A = \mathcal{M}(T,\mathcal{B})$ and $A' = \mathcal{M}(T,\mathcal{B}')$, and let P be the change-of-basis matrix of Definition 4.24. Then

$$A' = P^{-1}AP.$$

Proof. For any v, applying (4.2) and Theorem 4.6 repeatedly,

$$A'[v]_{\mathcal{B}'} = [Tv]_{\mathcal{B}'} = P^{-1}[Tv]_{\mathcal{B}} = P^{-1}A\,[v]_{\mathcal{B}} = P^{-1}AP\,[v]_{\mathcal{B}'}.$$

Since this holds for all coordinate vectors $[v]_{\mathcal{B}'}$, we conclude $A' = P^{-1}AP$ (Figure 2). ∎

$$\begin{tikzcd}
\mathbb{F}^n_{\mathcal{B}'} \arrow[r, "A' = P^{-1}AP"] \arrow[d, "P"'] & \mathbb{F}^n_{\mathcal{B}'} \arrow[d, "P"] \\
\mathbb{F}^n_{\mathcal{B}} \arrow[r, "A"'] & \mathbb{F}^n_{\mathcal{B}}
\end{tikzcd}$$

Figure 2: Similarity: to apply T in the new basis, convert to the old basis (P), apply A, and convert back (P^{-1}).

Definition 4.27 (Similar matrices). Square matrices A, A' are *similar* if $A' = P^{-1}AP$ for some invertible P. Similar matrices are exactly the matrices of one operator in different bases.

Proposition 4.28 (Similarity invariants). Similarity is an equivalence relation, and similar matrices have the same trace and the same determinant.

Proof. Reflexivity ($P = I$), symmetry ($A = PA'P^{-1}$), and transitivity (compose the P's) are routine. For the trace, the identity $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ gives $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}\big(A(PP^{-1})\big) = \operatorname{tr} A$. For the determinant, $\det(P^{-1}AP) = \det(P)^{-1}\det(A)\det(P) = \det A$. ∎

Trace and determinant are therefore properties of the *operator*, not of any particular matrix. The same will turn out to hold for the eigenvalues (Lecture 5): similar matrices share them, which is why diagonalizing is meaningful.

Remark 4.29 (Operators versus general maps). Similarity compares the two matrices of one *operator*, using the same basis on domain and codomain. For a general $T \in \mathcal{L}(V, W)$, with independent basis changes P on V and Q on W, the corresponding formula is $A' = Q^{-1}AP$, and such matrices are called *equivalent*. Equivalence is weaker: every rank-r map is equivalent to the block matrix with an $r \times r$ identity in the top-left corner and zeros elsewhere. Similarity preserves the finer data—trace, determinant, eigenvalues—that only makes sense when domain and codomain coincide.

Example 4.30 (A reflection diagonalized by a clever basis). Let $T \in \mathcal{L}(\mathbb{R}^2)$ be reflection across the line $y = x$, so $T(x, y) = (y, x)$. In the standard basis $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Choose the new basis $\mathcal{B}' = \big((1,1), (1,-1)\big)$, giving $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $P^{-1} = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. Then

$$A' = P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In the basis adapted to the reflection—along and across the mirror line—the matrix is diagonal, with entries the obvious scalings $+1$ and -1. This is a first glimpse of diagonalization: the right basis lays an operator bare. ◁

Example 4.31 (The same matrix, straight from the definition). We can also obtain $\mathcal{M}(T, \mathcal{B}')$ without the similarity formula. Since $T(1,1) = (1,1) = 1 \cdot (1,1) + 0 \cdot (1,-1)$ and $T(1,-1) = (-1,1) = 0 \cdot (1,1) + (-1) \cdot (1,-1)$, the columns are $(1,0)$ and $(0,-1)$, so $\mathcal{M}(T, \mathcal{B}') = \operatorname{diag}(1,-1)$ directly. The two routes agree—as Theorem 4.26 guarantees they must. ◁

Example 4.32 (Similarity links the two matrices of D). Return to D on $\mathcal{P}_2(\mathbb{R})$ in the standard basis $\mathcal{B} = (1, t, t^2)$ and in $\mathcal{B}' = (1,\ 1+t,\ 1+t+t^2)$ from Example 4.5. The change-of-basis matrix carries the $\mathcal{B}'$-vectors as columns in $\mathcal{B}$-coordinates:

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

With $A = \mathcal{M}(D, \mathcal{B}) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, a short multiplication gives

$$P^{-1}AP = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D, \mathcal{B}'),$$

matching the direct computation of Example 4.5, exactly as Theorem 4.26 demands. ◁

Example 4.33 (Invariants in action). For the reflection of Example 4.30, $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has $\operatorname{tr} A = 0$ and $\det A = -1$, while the diagonalized $A' = \operatorname{diag}(1,-1)$ has $\operatorname{tr} A' = 1 + (-1) = 0$ and $\det A' = -1$. The invariants agree, as Proposition 4.28 promises—and we can already anticipate that this reflection's eigenvalues are $+1$ and -1, the diagonal entries laid bare by the adapted basis. ◁

The map, not the matrix

A matrix is a basis-dependent shadow of a linear map. Quantities that survive a change of basis—rank, trace, determinant, the characteristic and minimal polynomials, eigenvalues, and later canonical-form data—belong to the operator itself. Individual entries and coordinate patterns generally do not. The program of the next several lectures is to exploit this freedom: to hunt for a basis in which the matrix is diagonal, so that the operator's action becomes transparent. In a finite-dimensional or discrete pure-point quantum model, an energy eigenbasis makes $e^{-i\hat{H}t/\hbar}$ diagonal and the dynamics decouples into independent phases. Lectures 13–14 replace this ordinary matrix picture by generalized/spectral representations when continuous spectrum is present.

Remark 4.34 (Forward preview: density matrices). In the later finite quantum model, a density matrix ρ is a positive trace-one operator representing a state, and a self-adjoint operator A represents an observable. Under the same basis change both transform simultaneously: $\rho' = P^{-1}\rho P$ and $A' = P^{-1}AP$. Therefore

$$\operatorname{tr}(\rho' A') = \operatorname{tr}(P^{-1}\rho A P) = \operatorname{tr}(\rho A).$$

Thus the expectation $\langle A\rangle = \operatorname{tr}(\rho A)$ is independent of the chosen coordinates. Lecture 10 supplies the physical definitions and hypotheses; this is only a preview of the change-of-basis calculation.

Summary of main concepts

Fixing bases of V and W turns each $T \in \mathcal{L}(V,W)$ into a matrix $\mathcal{M}(T)$ whose columns are the images of the basis vectors in coordinates, and $[Tv]_{\mathcal{C}} = \mathcal{M}(T)[v]_{\mathcal{B}}$. Composition of maps becomes multiplication of matrices—which is what forces the row-by-column rule—so $\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$ and $\dim \mathcal{L}(V,W) = (\dim V)(\dim W)$. A map is an isomorphism iff it is bijective, and finite-dimensional spaces are isomorphic iff their dimensions agree, so every n-dimensional space is a copy of $\mathbb{F}^n$. Changing basis replaces an operator's matrix A by the similar matrix $P^{-1}AP$; trace and determinant are similarity invariants and hence belong to the operator.

- **Matrix of a map**—its columns are the images of the basis vectors in coordinates; basis-dependent.
- **Composition → multiplication**—forces the row-by-column rule; $\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$.
- **Isomorphism**—a bijective linear map; finite-dimensional spaces are isomorphic $\iff$ their dimensions agree.
- **Change of basis / similarity**—$A \mapsto P^{-1}AP$; trace and determinant are invariant, so they belong to the operator.

Exercises for this lecture are in the companion sheet `exercises-4.pdf`.